TTC antidiagonal interpretation

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Focusing **only on anti-diagonal lineouts** (fixed T, varying τ) is exactly the right move here. That isolates a single “age” of the system and asks how correlations behave symmetrically before and after that time.

Let’s be very precise about what an **oscillation** along the anti-diagonal can and cannot mean.

1. What an anti-diagonal lineout really measures

Along the anti-diagonal through $t_1=t_2=T$:

$(t_1, t_2) = (T+\Delta,\, T-\Delta) \quad\Rightarrow\quad \tau = -2\Delta$

So an anti-diagonal lineout is effectively:

$C(T,\tau)$

at **fixed experimental time** T.

This is the cleanest possible measurement of **intrinsic temporal structure**, uncontaminated by aging or drift.

2. What oscillations cannot be (important eliminations)

An oscillation along the anti-diagonal **cannot** be due to:

❌ Aging

Aging lives in T. You have fixed T.

❌ Slow drift (temperature, beam, alignment)

Drift breaks symmetry between t_1 and t_2. Anti-diagonal is explicitly symmetric.

❌ Simple diffusive relaxation

Diffusion produces monotonic decay in τ, never oscillations.

So if oscillations are real, they are **not a trivial artifact of non-stationarity**.

3. What oscillations do mean physically

An oscillation in $C(T,\tau)$ implies:

The system “remembers” past configurations in a phase-coherent way.

That already narrows the interpretation enormously.

Below are the physically meaningful possibilities, ordered from most to least plausible in Bragg-XPCS contexts.

4. Most likely: periodic or quasi-periodic internal dynamics

Physical interpretation of oscillations in anti-diagonal TTC lineouts

An anti-diagonal lineout through a two-time correlation (TTC) map corresponds to fixing the average experimental time $T = (t_1+t_2)/2$ and varying the time separation $\tau = t_2 - t_1$. In this representation, the system is probed symmetrically before and after a given experimental time, and any evolution of the correlation function reflects intrinsic temporal structure rather than aging or slow drift.

An oscillatory signal along the anti-diagonal therefore cannot be attributed to aging, instrumental drift, or simple diffusive relaxation. Aging manifests as a dependence on T, which is fixed along the anti-diagonal, while drift breaks the symmetry between t_1 and t_2. Purely diffusive dynamics produce monotonic decay in τ and cannot generate oscillations. Consequently, oscillations in anti-diagonal lineouts indicate genuinely non-diffusive dynamics.

Physically, such oscillations imply that the system revisits similar microscopic configurations in a phase-coherent manner. This behavior is consistent with periodic or quasi-periodic internal dynamics, in which certain degrees of freedom undergo recurrent motion. Examples include grain-boundary sliding, oscillatory strain relaxation, domain-wall motion pinned between defects, or collective rearrangements of ions within constrained environments. In these cases, the correlation function at fixed T may take the form of a damped oscillation in τ, reflecting reversible or underdamped motion rather than irreversible relaxation.

Another possible origin is periodic external driving that is not directly apparent in the raw intensity time series but becomes visible through correlations. Weak periodic forcing from temperature regulation, pressure control, mechanical vibrations, or beamline feedback systems can imprint a characteristic frequency on the TTC. Such effects typically produce oscillations with a well-defined and time-independent period across different regions of reciprocal space.

Oscillatory anti-diagonal correlations may also arise from intermittent switching between metastable states. In systems with multiple local energy minima, transitions between configurations can be temporally structured rather than purely stochastic. Following a transition, the system may exhibit a partial recovery or “ringing” behavior, leading to oscillatory correlations. In this scenario, oscillations may be mask-dependent in phase or amplitude, reflecting spatial heterogeneity in the underlying dynamics.

In Bragg XPCS, these effects are further amplified by the phase sensitivity of coherent scattering. Small, coherent displacements or strain variations can strongly modulate the speckle pattern, meaning that reversible motions on very small length scales can generate pronounced oscillatory features in the TTC.

To distinguish between these scenarios, several diagnostic tests are informative. Comparing oscillation frequencies and phases across different reciprocal-space masks can reveal whether the dynamics are global or spatially localized. Tracking the oscillation period as a function of experimental time T can separate steady external driving from evolving internal dynamics. Verifying symmetry of the correlation function for positive and negative τ confirms that the signal is genuinely anti-diagonal and not contaminated by aging. Finally, comparing anti-diagonal lineouts with cuts parallel to the diagonal can clarify whether oscillatory behavior is intrinsic or coupled to slow evolution.

Overall, oscillations observed along the anti-diagonal of a TTC map indicate that the system does not undergo simple relaxational dynamics but instead exhibits temporally structured, partially reversible microscopic processes. This constitutes strong evidence for non-diffusive, collective, or intermittently driven dynamics in the probed material.

Physical picture

Some internal degree of freedom undergoes **recurrent motion**:

- hopping back and forth
- switching between configurations
- elastic relaxation followed by recovery
- collective modes in confined regions

If the system partially revisits similar configurations after a time τ_0, then:

$$C(T,\tau) \sim \cos(\omega \tau)\, e^{-\tau/\tau_c}$$

with:

- oscillation period $2\pi/\omega$
- damping time τ_c

Concrete examples

- grain-boundary sliding that repeats
- domain wall oscillations pinned between defects
- strain relaxation and rebound
- ionic rearrangements in a constrained cage

This is **not** normal diffusion. It is **underdamped or quasi-reversible motion**.

5. Periodic driving hidden inside the experiment

Sometimes oscillations reflect an **external clock** that is invisible in the time trace but visible in correlations:

- cryostat temperature regulation
- pressure regulation
- beamline feedback loops
- mechanical vibrations

Why this shows up in anti-diagonal:

- the TTC integrates over *pairs* of times
- even weak periodic forcing becomes coherent in $C(T,\tau)$

Rule of thumb:

- If the oscillation period is **exactly constant** over many T, be suspicious of instrumentation.
- If it drifts slowly or appears only at specific T, it is likely intrinsic.

6. Intermittent switching between metastable states

This is extremely relevant to **ionic conductors and complex solids**.

Imagine:

- two or more structural microstates
- switching is not random but has preferred timescales
- system “rings” after a transition

Then:

- anti-diagonal lineout shows oscillations
- diagonal-parallel cuts show checkerboard or blocky TTC textures
- different masks show oscillations at different phases or amplitudes

This matches **mask-selective oscillations**, which you’ve already observed.

7. Collective modes in Bragg XPCS (important nuance)

In Bragg geometry, you are not measuring particle positions but **phase-sensitive strain or displacement fields**.

That means:

- small coherent shifts can strongly affect speckle
- elastic modes can produce oscillatory correlations
- even Å-scale reversible motion is visible

So oscillations do **not** require large atomic motion.

8. How to tell which scenario you're in (practical tests)

A. Compare different masks

- Same frequency everywhere → global mode or external driver
- Mask-dependent phase or frequency → local dynamics

B. Track frequency vs T

- Constant → instrumental or steady collective mode
- Slowly changing → evolving microstructure

C. Check symmetry

True anti-diagonal oscillations must be:
C(T,+\tau) = C(T,-\tau)
If not, you're mixing in aging or drift.

D. Compare with diagonal-parallel cuts

- Oscillation only in anti-diagonal → intrinsic dynamics
- Oscillation also in diagonal-parallel → aging + recurrence

9. A key conceptual takeaway

An oscillation along the anti-diagonal means the system is not simply relaxing — it is dynamically structured in time.

That is a *strong* statement.

In many XPCS papers, seeing this would already justify:

- "non-diffusive dynamics"
- "reversible or quasi-elastic rearrangements"
- "collective modes"
- or "periodic/intermittent microscopic processes"

10. Why referees will care

Anti-diagonal oscillations:

- are immune to most trivial artifacts
- are rarely shown
- strongly support claims of non-equilibrium or complex dynamics

Handled carefully, this is **a central result**, not a side observation.

If you want, next we can:

- map oscillation period → physical length scales
- build a minimal toy model that reproduces your TTC
- or discuss how to present this convincingly for PRL/Nature-style audiences